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Leadership going social: How U.S. nonprofit executives engage publics on Twitter

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ABSTRACT

This study explored how nonprofit executives in the U.S. engaged with online publics through various communication strategies. We investigated the extent to which nonprofit executives leveraged 1) digital dialogic principles, 2) social presence strategies, and 3) message tactics. In addition, this study attempted to explore how different communication strategies influenced publics' digital engagement with nonprofit leaders. We conducted a content analysis on 700 Twitter posts from 35 nonprofit executive leaders who possess an active Twitter presence and 195 public comments. These executives led America's largest nonprofits across a wide range of religious, educational, scientific, sports, and social welfare organizations. The results showed that nonprofit executives employed some degree of the four dialogic principles and capitalized on each of the three types of social presence strategies evenly. Regarding message function, nonprofit executives were most active in disseminating information and creating an online community. Findings regarding the associations between communication strategies and public engagement remain mixed.

Throughout history, nonprofit organizations have played a critical role in fostering a vibrant democratic society (Norris-Tirrell et al., 2018). Specifically, they fulfill essential social functions for modern societies by providing resources, services, and goods while also spotlighting unaddressed social problems, mobilizing collective actions, and driving civic engagement (Payton and Moody, 2008). The nonprofit sector in the United States continues to expand, contributing to 5.6 percent of the country's GDP in 2016 (National Center for Charitable Statistics, 2016). There were 1.54 million registered nonprofits in the U.S. comprising a wide range of programs and services in the arts, health, education, spirituality, social services, and business, to name a few (National Center for Charitable Statistics, 2016).

Over the past few years, large and mid-sized nonprofits have rapidly adopted social media to communicate with internal and external constituents (Guo and Saxton, 2014). Social media proves to be an inexpensive resource for nonprofits, playing a pivotal role in accumulating social capital, enhancing visibility, strengthening stakeholder engagement and building relationships (Svensson et al., 2015; Xu and Saxton, 2019). Not surprisingly, the adoption and effectiveness of social media by nonprofits has been at the center of communication and nonprofit research over the last decade (e.g., Auger, 2014; Bortree and Seltzer, 2009; Lovejoy and Saxton, 2012; Wang and Yang, 2020). However, this interest has mainly focused on organizational-level communication (i.e., analyzing

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organizations' official websites or social media accounts). In comparison, there is a dearth of research concerning social media communication by nonprofit executives who often lack more traditional resourcing, organizational, or structural support compared to their for-profit counterparts (Dolan, 2002; Waters, 2014).

Nonprofits are led by paid or unpaid executive leaders, more commonly referred to as Executive Director, Chief Executive Officer, or President (Norris-Tirrell et al., 2018). As noted by Kearns et al. (2015), most nonprofit CEOs are at the epicenter of their organization: They have substantial authority and assume primary responsibility for their nonprofit's success. In examining skills required for nonprofit leaders to perform leadership tasks, Kearns et al. (2015) identified interpersonal skills, communication, and trust-building as being essential to leadership performance and credibility. Leaders who possess exceptional attention to detail, external communication skills, boundary-spanning, and relationship management capability ultimately contribute to a nonprofit's success (Carlson and Donohoe, 2010; Dollinger, 1984). Despite a growing body of research examining nonprofit leadership skills, styles, characteristics, motivation, leader development, and executive transition and succession (e.g., Hodges and Howieson, 2017; Kuenzi and Stewart, 2017; Norris-Tirrell et al., 2018; Phipps and Burbach, 2010), there is a surprising dearth of information concerning how nonprofit leaders communicate with publics on social media.

Given the immense potential of nonprofit executives as key relationship builders and the lack of research examining nonprofit executives' communication on social media, this study explores how U.S. nonprofit executives engaged with online publics via communication strategies and tactics. Specifically, we conducted a content analysis on 700 Twitter posts by 35 nonprofit executive leaders who possess an active Twitter presence and 195 public comments. Drawing literature primarily from dialogic communication, social presence, and the information-communication-action (ICA) framework, we investigated the extent to which nonprofit executives leveraged 1) digital dialogic principles, 2) social presence strategies, and 3) message tactics. This study also explored how different communication strategies and tactics influenced publics' digital engagement with nonprofit leaders.

This empirical investigation into nonprofit leadership communication has important theoretical implications for research in nonprofit communication and management, leadership communication, and social media communication. This study also benefits communication professionals and executive leaders working in nonprofit organizations by suggesting research-informed communication strategies and tactics tailored as necessary for specific executive leaders.

1. Literature review

1.1. Nonprofit and public relations

Nonprofits rely on external sources for both resources and legitimacy (Brown, 2005; Ospina et al., 2002). To survive and thrive, nonprofits must cultivate a group of committed stakeholders who can provide support in the form of funding, labor, and advocacy (Carlson and Donohoe, 2010). The long list of stakeholders includes clients, prospective funders, community leaders, politicians, media, regulatory agencies, volunteers, and employees, all of which impact nonprofits' operational environment and strategic effectiveness (Balser and McClusky, 2005; Carlson and Donohoe, 2010; Doerfel et al., 2017). Therefore, being responsive to stakeholder needs, maintaining stakeholder relationships, and aligning stakeholders' expectations with respect to good governance remain essential ingredients for effective nonprofits (Balser and McClusky, 2005; Wellens and Jegers, 2014). As Ospina et al. (2002) noted, successful nonprofits are capable of coordinating stakeholders' diverse needs and interests and addressing competing pressures from stakeholders to create win-win solutions.

Public relations and marketing have long been viewed as crucial functions through which nonprofits enhance visibility, impact, and loyalty among key stakeholders (Pressgrove and McKeever, 2016). Two decades of empirical research shows that effective public relations activities—such as communicating nonprofits' mission, values, activities, and achievements while also using dialogue to address questions and needs of the community—has contributed to robust donor/volunteer relationships, donor engagement, positive word-of-mouth, and donation behaviors (e.g., Harrison et al., 2017; Kelly, 1998; O'Neil, 2009; Waters, 2011). With the development of digital technology, the function of public relations in nonprofit organizations has evolved and extended beyond traditional printed media to now encompass computer-mediated communication (Bortree and Seltzer, 2009; Saxton and Waters, 2014).

1.2. Nonprofits and social media communication

Contemporary nonprofit organizations consider social media as a relatively low-cost means of reaching large and geographically dispersed stakeholders and enhancing organizational visibility (Auger, 2014; Cho et al., 2014; Lovejoy and Saxton, 2012; Maxwell and Carboni, 2016). Indeed, the increasingly interconnected environment nonprofits operate in to bridge intersectoral resources makes it imperative that nonprofits utilize the interactive function of social media to branch out and build social capital (Xu and Saxton, 2019). Social media offers a positive, diverse, and transcendent public sphere that can spur public dialogue and facilitate nonprofit stakeholder engagement, relationship building, advocacy, knowledge acquisition, and volunteer management (Farrow and Yuan, 2011; Guo and Saxton, 2018; Patel and McKeever, 2014; Svensson et al., 2015). Over the past decade, a sizable body of research has empirically explored how various types of nonprofits leverage websites and social media to communicate, engage, and build relationships with their stakeholders (e.g., Bortree and Seltzer, 2009; Guo and Saxton, 2014; Qu, 2020; Saxton and Waters, 2014; Xu and Saxton, 2019). For instance, focusing on U.S. nonprofits, Saxton and Waters (2014) found that publics prefer organizational efforts aimed at building a sense of community and dialogue over one-way information. In particular, call-to-action messages were highly effective in eliciting publics' likes and comments. Wang and Yang (2020) studied U.S. nonprofits' use of dialogic communication and concluded that providing useful, regularly updated information was the major function of nonprofit organizations' Twitter use, which benefitted

nonprofits in attracting retweets and likes. Additionally, a perusal of literature reveals that two theoretical frameworks primarily inform empirical assessments on this topic: (1) dialogic communication (Kent and Taylor, 1998) and (2) the information, community-building, and action (ICA) framework (Lovejoy and Saxton, 2012).

Despite the burgeoning body of knowledge on nonprofit social media use, how nonprofit leaders directly engage with publics via social media is rarely mentioned. Nonprofit executives play a vital part in connecting organizations with external stakeholders (Carlson and Donohoe, 2010). Through personal branding and additional opportunities for engagement, leadership communication typically functions as a more human-side of organizational communication to authenticate or personalize the otherwise mundane, impersonal organizational-level communication (e.g., Tsai and Men, 2017; Yue et al., 2019). Below, we describe in detail the roles and functions of nonprofit leaders and their communication behavior.

2. Nonprofit leadership

2.1. Roles and functions

As expectations of the nonprofit sector have remarkably increased, leaders of nonprofits are expected to engage in more active stakeholder dialogue, identify and collaborate with an increasing number of stakeholders, and use their power in an ethical manner (Hodges and Howieson, 2017). Compared to their for-profit counterparts, nonprofit leaders face unique challenges on multiple fronts, such as fundraising, staffing, volunteer retention, and increased accountability demands (Dolan, 2002; Waters, 2014). There are growing expectations for nonprofit executives to look beyond their organization to become more externally focused, such as monitoring the changing external environment, attending various fundraising events, adopting collaborative ways of working across sectors, and speaking directly with donors, government officials, community members, and media (Hodges and Howieson, 2017; Ingram, 2019). Nonprofit leaders are also expected to serve in a more entrepreneurial role as compared to the typical chief executive in the corporate world (Taliento and Silverman, 2005). As Hodges and Howieson (2017) pointed out, nonprofit leaders need not only to improve organizational performance and effectiveness. More importantly, they should cultivate the meaning, morality, and culture of their nonprofit. Experienced nonprofit leaders are also proficient in communicating with responsiveness and building robust engagement with various stakeholders by cultivating long-term strategic partnerships (Balser and McClusky, 2005; Herman, 2016). There remains a long list of leadership characteristics required within the nonprofit sector, with one common theme highlighting the importance of a communicative 'ambassadorial' dimension in leadership, as leaders assume key roles in networking, representation, vision articulation, and conversation (Hodges and Howieson, 2017; Peck et al., 2009).

2.2. Executive leadership communication

Leadership and communication are inherently linked. Communication is not only a valuable skill that complements the leadership repertoire but is the central, fundamental element that constitutes and constructs leadership (Fairhurst and Connaughton, 2014). Nevertheless, leadership communication is often considered "the elephant in the room of leadership" (Mayfield and Mayfield, 2017, p. 6) as most research collapses assessments of leader communication into broader leadership styles, such as charismatic-transformational leadership (van Knippenberg and Stam, 2014). This study dives into the precise communication strategies utilized by nonprofit executives on social media by embracing the conceptual richness of leader communication behavior. This focus helps us explore which leadership communication strategies are more likely to elicit positive outcomes in a computer-mediated environment, thus generating strategic value to executive leaders and their communication teams in the nonprofit sector.

Communication and public relations scholars have recently spearheaded efforts in establishing the role of executive leaders as engagement officers and relationship builders (e.g., Alghawi et al., 2014; Huang and Yeo, 2018; Men et al., 2018; Tsai and Men, 2017; Yue et al., 2020). Previous research outlines how senior corporate leaders utilize social media to communicate with online stakeholders on behalf of their organizations (Tsai and Men, 2017; Men et al., 2018; Yue et al., 2019). These studies examined CEOs' adoption of dialogic communication, social presence strategies, personal and corporate disclosure, and various message tactics. More importantly, they revealed the impact of different communication tactics on individual and organizational outcomes, such as public engagement with organizations, organization-public relationship building, and leaders' personal branding results (e.g., Huang and Yeo, 2018; Kim and Sung, 2020; Men et al., 2018; Tsai and Men, 2017; Yue et al., 2020). While this line of research provides considerable insights into leadership communication practices in the computer-mediated context, they predominantly spotlighted *corporate* leaders ranging from Fortune CEOs (Huang and Yeo, 2018; Yue et al., 2019), top startup CEOs (Yue et al., 2019), and top social CEOs in the for-profit sector (Men et al., 2018). The current study presents one of the first comprehensive investigations into the nature, extent, and impact of *nonprofit* leader communication on social media. Below we reviewed key literature in dialogic communication (Kent and Taylor, 1998) and the ICA framework (Lovejoy and Saxton, 2012) to inform our research questions.

2.3. Leaders' communication strategies on social media: dialogic communication

Kent and Taylor's (1998) conceptualization of dialogic communication has spawned numerous discussions, refinements, and applications in the field of public relations over the past 20 years (Morehouse and Saffer, 2018; Zhou and Xu, 2019). In essence, dialogue refers to a relational orientation between organizations and stakeholders encompassing five features: mutuality (i.e., emphasizing collaboration and a spirit of mutual equality), propinquity (i.e., addressing dialogic exchanges and mutual consultation), empathy (i.e., an atmosphere of trust and support for dialogue to succeed), risk (i.e., taking relational risks in dialogic communication),

and commitment (i.e., embracing genuineness, authenticity, and a commitment to conversations and interpretations) (Kent and Taylor, 2002). As an ethical practice, dialogic communication signals the co-creational turn in public relations aimed at building mutual understanding, social capital, and long-term organization-public relationships desirable for organizations, publics, and the society (Kent and Lane, 2017; Taylor and Kent, 2014).

2.3.1. The functional dimension of dialogic communication: Digital dialogic principles

The five digital dialogic principles are a set of practical suggestions proposed by Kent and Taylor (1998) to guide organizations in building dialogue in the specific settings of digital technologies. The five principles include (1) creating dialogic loops, which emphasizes organizational listening, soliciting feedback, and answering questions from their publics; (2) providing useful, relevant information of interest to their publics; (3) generating return visitors by incorporating dynamic features into the sites; (4) conserving visitors by providing links to organizational sites; and (5) designing a user-friendly and easy-to-navigate interface (Kent and Taylor, 1998). The five principles should not be conflated with the dialogic communication theory. Instead, they act as the digital manifestation of dialogic communication and should be considered as procedural steps that may ultimately fulfill the dialogic promise (Taylor and Kent, 2014; Sommerfeldt and Yang, 2018; Zhou and Xu, 2019).

Public relations scholars have adapted and applied the five principles in a vast amount of studies across different organizational websites and social media platforms (Sommerfeldt and Yang, 2018). Its popularity is driven by scholars' keen interest in digital technologies' utility for building organization-public relationships and public engagement. Research in this vein has produced consistent findings that organizations, including nonprofits, failed to utilize social media's dialogic potential fully (e.g., Bortree and Seltzer, 2009; Qu, 2020; Rybalko and Seltzer, 2010; Wang and Yang, 2020). In comparison, preliminary evidence shows that the interpersonal nature of *leadership* communication could facilitate the implementation of dialogic communication principles (e.g., Men et al., 2018).

Traditional operationalization of dialogic communication is primarily functional and does not reflect the interpersonal and relational nature of dialogue initiated by organizations. Criticism of the five principles surrounds rigidly reifying dialogue as a set of procedures and categories resembling online advertising and product promotion as opposed to genuine dialogue (Kent and Lane, 2017; Taylor and Kent, 2014). In other words, the relational features of dialogic communication are not fully captured in the five digital dialogic principles (Men et al., 2018; Saxton and Waters, 2014). Scholars agree that adopting certain digital media features does not automatically generate dialogue or enhance public engagement. Rather, it is through carrying the dialogic orientation and implementing the core principles of dialogue and dialogic communication (i.e., mutuality, propinquity, empathy, risk, commitment) for an organization to fulfill its dialogic promise.

In recent years, the original proponents of dialogic communication have called for re-examining the philosophical roots of dialogue (Kent and Lane, 2017; Kent and Theunissen, 2016; Taylor and Kent, 2014). Echoing the theory's foundation in interpersonal communication and relational orientation, we follow recent efforts to incorporate *social presence* into the operationalization of digital dialogic communication (cf. Men et al., 2018).

2.3.2. The relational dimension of dialogic communication: social presence strategies

Social presence remains one of the most studied constructs in exploring interpersonal interactions and relationships in the context of computer-mediated communication (CMC) (Walther, 2011). A classic definition of social presence is "the degree of salience of the other person in the interaction and the consequent salience of the interpersonal relationships (Short et al., 1976, p. 65). Social presence is often operationalized as the degree to which a medium or a mediated partner is perceived as personal, real, accessible, sociable, or intimate (Lombard and Ditton, 1997). Over time, scholars found that social presence is less reliant on the medium's features/attributes as compared to the properties of communication interaction (Biocca et al., 2003; Sivunen and Nordbäck, 2015).

In studying social media-based communication, researchers have contextualized social presence as "the extent to which a person feels as if he/she were 'with' the communication partner, engaging in a direct, face-to-face conversation" (Lee and Shin, 2012, p. 516). This definition emphasizes the sense of being with another in a mediated setting (Biocca et al., 2003) manifested through perceived intimacy and immediacy between the interactants (Rice, 1993). Social presence is fundamental to building interpersonal relationships in the CMC context (Kietzmann et al., 2011). Numerous studies show that the psychological involvement inherent in social presence is essential to establishing a sense of mutuality, common ground, connections, and relationships (Biocca et al., 2003). For instance, public figures' strong social presence on Twitter resulted in a more positive evaluation of public figures and behavioral intention to support these same public figures (Lee and Jang, 2013). Even in the complete absence of face-to-face interactions, social presence also forged audiences' parasocial relationships (i.e., illusory personal relationships) with media personalities (Thorson and Rodgers, 2006).

Social presence has interested scholars in public relations and leadership communication for the last decade (Kelleher and Miller, 2006; Men et al., 2018; Oh and Ki, 2019; Park and Cameron, 2014). Social presence strategies adopted online by organizations or leaders of organizations have contributed to organizational effectiveness in several ways, resulting in enhanced organizational reputation (e. g., Park and Cameron, 2014), organization-public relationships (e.g., Kelleher and Miller, 2006), online public engagement (e.g., Yue et al., 2019), the persuasiveness of organizational messages (e.g., Lim and Lee-Won, 2017), and publics' positive attitude and behavioral advocacy toward organizations (e.g., Lim and Lee-Won, 2017; Park and Cameron, 2014).

In the present study, we examined nonprofit leaders' employment of three social presence strategies—affective, interactive, and cohesive strategies—borrowed from Rourke et al. (1999). First, affective strategy is concerned with leaders' expressing emotions, feelings, and detailing personal life and opinions to establish interpersonal connection and rapport. This dimension corresponds with the *risk* and *empathy* aspects of dialogic communication, which necessitates the disclosure of vulnerability and demonstration of trust and support in building dialogue (Kent and Taylor, 2002). Second, interactive strategy entails addressing the needs of other

communication partners by answering their questions, recognizing their feedback, and showing appreciation and agreement. Such an approach aligns with the *propinquity* and *mutuality* principles of dialogic communication in highlighting dialogue as collaboration, reciprocity, mutual consultation and equity (Zhou and Xu, 2019). Finally, cohesive strategy refers to building a sense of community by using inclusive pronouns, vocatives, and salutations. Therefore, it relates to the *commitment* and *mutuality* aspects of dialogue, in which a sense of togetherness, interdependence, and relationship commitment is captured. Overall, the three communicative strategies complement the existing digital dialogic principles by highlighting the social, relational, and interpersonal orientation of dialogue originally envisioned by Kent and Taylor (2002). Additionally, incorporating the relational dimension is particularly relevant to research mediated leadership communication, as senior leaders, in contrast to anonymous representatives operating corporate social media accounts, are expected to be more conversational, human, and animated (Men et al., 2018).

We selected Twitter as the social media platform to conduct the content analysis because Twitter remains one of the most frequently used social media platforms by CEOs (CEO.com and DOMO, 2016; Yue et al., 2019). Compared to other platforms such as Facebook, whose primary purpose is to connect individuals with their close network, and LinkedIn, which is more professionally-oriented, Twitter is largely used for wider opinion exchange and organization-public communication. In addition, we identified an adequate number of nonprofit leaders who were active on Twitter, which will be further addressed in the method section. Thus, we consider Twitter as a suitable platform to examine nonprofit executives' communication and engagement with publics (Forsey, 2020; Huang and Yeo, 2018). Our first research question focuses on leaders' use of dialogic communication on Twitter.

RQ1: How, if at all, is dialogic communication (i.e., dialogic principles and social presence strategies) employed in nonprofit executives' Twitter posts?

2.4. Message function and message appeal on social media

2.4.1. Message function.

Lovejoy and Saxton (2012) proposed the Information-Community-Action (ICA) framework in their investigation of message functions by nonprofit organizations on social media. We believe that incorporating the ICA framework into the current study provides additional insights into communication interests specific to nonprofit leaders. Even though dialogic communication framework offers valuable insights regarding the informational and dialogic properties of social media messages, it does not capture message functions aimed at mobilizing public action and building online communities. Nevertheless, for many nonprofit organizations and leaders, their primary goal is to promote nonprofit causes and mobilize people's actions rather than to create dialogue (Lovejoy and Saxton, 2012). In other words, the action function, which involves asking followers to act on something concrete, is often what many nonprofits ultimately want from their social media operation (Lovejoy and Saxton, 2012; Saxton and Waters, 2014). Therefore, despite a certain degree of overlap with dialogic communication, we argue that the benefits of examining message functions outweigh concerns in that it deepens our understanding of nonprofit leaders' role in promoting actions and building online communities.

To elaborate on the framework, *information function* is a one-way messaging strategy that primarily serves to inform publics about organizational activities, events, news, facts, and updates. Conceptually, it is equivalent to the "providing information of interest to stakeholders" principle of functional digital dialogic communication. Second, *community function* entails genuine outreach to publics to create online communities and maintain good relationships (Saxton and Waters, 2014). Similar to the "dialogic loop" concept, community-building messages may involve direct conversations with publics (e.g., responding to publics' messages, soliciting responses); however, not all messages need to be interactive to fulfill the purpose of building and strengthening ties to the online community (Lovejoy and Saxton, 2012). For instance, organizations may simply reference social issues they are involved with or mention work performed by their competitors or partners with no intention to generate conversations. Lastly, *action function* focuses on promotion and mobilization. Although one-way in nature, these messages encourage people to engage in supporting behaviors, including making donations, lobbying and advocacy, promoting or participating in events, and learning how to help.

2.4.2. Message appeal

This study strived to understand different message appeals used by nonprofit leaders to provide a more holistic view of leaders' social media communication. Message appeals, whether rational or emotional, are frequently used by organizations as communication strategies across various channels (Ji et al., 2019; Liu et al., 2017). Literature has extensively investigated message appeals and its persuasive effects on individual attitudes and behaviors (Lwin and Phau, 2013). For instance, message appeals could influence an individual's engagement with organizations' social media posts such that emotional appeals found in Fortune 500 companies' Facebook posts led to more likes, shares, and comments than informational posts (Liu et al., 2017).

We applied the rational/emotional framework to examine different message appeals (Albers-Miller and Stafford, 1999). Rational appeals involve using compelling and logical arguments to persuade the target audience, whereas emotional appeals aim to arouse audiences' emotional responses (Albers-Miller and Stafford, 1999). More specifically, Stafford and Day (1995) referred to rational appeals as the direct, straightforward, and objective presentation of factual information. Emotional appeals, on the other hand, are used to generate either positive (e.g., joy, love, humor, pride) or negative feelings (e.g., guilt, shame, anger) from the target audience (Zhang et al., 2014). The effectiveness of the two appeals often varies across communication contexts (Liu et al., 2017). As communication scholar Nabi (2007) noted, it is essential to examine both elements to find the right combination.

Overall, our second research question explores different message functions and message appeals used by nonprofit executives.

RQ2: How, if at all, are different message functions and message appeals employed by nonprofit executives on Twitter?

2.5. Social media engagement

Research on social media engagement exponentially increased as public relations scholars began to recognize the indispensable role of social media in building public engagement and organization-public relationships (Men and Tsai, 2014; Morehouse and Saffer, 2018; Yue et al., 2020; Zhou and Xu, 2019). Social media engagement signals publics' proactivity and openness in participating in conversations with organizations. In other words, the varying levels of social media engagement reflect the willingness and commitment of publics in consulting and negotiating with organizations to reach mutually beneficial decisions. In addition, significant advancement in the field of social media engagement is attributed to the readily available, naturally occurring behavioral data manifesting through the number of likes, shares, and comments (Ji et al., 2019; Saxton and Waters, 2014). A perusal of literature indicates the predominance of such operationalization exists in evaluating publics' digital engagement with organizations and leaders (e.g., Cho et al., 2014; Ji et al., 2017; Liu et al., 2017; Men and Tsai, 2014; Saxton and Waters, 2014; Wang and Yang, 2020; Yue et al., 2019; Yue et al., 2020).

Consistent with prior literature, this study examines three behavioral indicators to quantify social media engagement with nonprofit leaders. The "Like" button signals audiences' interest or positivity toward the message (Swani et al., 2013). Posts liked by users likely appear in their followers' feeds. "Sharing" on Twitter can be done either through one direct click or quote retweet (i.e., a retweet that has been made with a comment). Compared to shares and likes, commenting on posts requires more cognitive effort by users. Notably, the number of comments is not equivalent to users' positive sentiment toward the post. Therefore, in the current study, we also evaluated the tone of top user comments and classified these comments into positive, neutral, and negative valence (Ji et al., 2017; Men et al., 2018). To explore publics' social media engagement with nonprofit executives, we ask the following research question.

RQ3: How do publics engage with nonprofit executives' Twitter posts in terms of likes, retweets, comments, and the tone of their comments?

Although previous studies have generally claimed the positive role of digital dialogic principle and ICA messages in generating publics' social media engagement (Ji et al., 2017; Rybalko and Seltzer, 2010; Wang and Yang, 2020), findings were mixed regarding the effectiveness of certain sub-principles and functions (Men et al., 2018; Thelen et al., 2021; Yue et al., 2019). For instance, there was evidence that "generation of return visits" decreased the number of user shares (Men et al., 2018) and likes (Thelen, et al., 2021). Similarly, another study found that each dialogic principle negatively affected different aspects of user engagement, with the exception of "posting information of interest to stakeholders" (Yue et al., 2019). Regarding the engagement function of ICA messages, there were also mixed results. For example, despite the overwhelmingly positive role of community-building messages, action-focused messages generated fewer responses from publics (Saxton and Waters, 2014). As a further case in point, Huang et al. (2016) found that community-building messages were not different from informational messages in eliciting the number of likes. Considering these mixtures of findings from prior research and the exploratory nature of the current study focusing on nonprofit leaders, we believe that it is more appropriate to propose a research question as opposed to a directional hypothesis, which demands consistent findings from previous studies. Therefore, we ask:

RQ4: How do dialogic communication, message function, and message appeal affect public engagement on Twitter?

2.6. Method

2.6.1. Sampling

As in prior studies (e.g., Saxton and Waters, 2014), this study adopted the renowned *Nonprofit Times 100* list published in 2019, which includes 100 of America's largest nonprofits based on total revenue[1]. Large nonprofits have more financial capacity to invest in social media management, therefore, it is reasonable to assume that leaders of large-sized nonprofits are likely to practice strategic social media communication with stakeholders. The primary nonprofit industries examined in this research focused on organizations purposed for religious, educational, charitable, scientific, public safety, amateur sports competition or prevention of cruelty to children and animals. These industry types represent the most frequent classification of nonprofit organizations, commonly coded as 501 (c)(3), which dictates eligibility, lobbying, and funding tied to private charitable contributions, grantmaking, and the tax code regulations.

We then identified 35 executive leaders (i.e., CEOs, presidents, or executive directors) from the list who have been active on Twitter[2]. Next, we verified the authenticity of these top leaders based on their profile descriptions, mentions by other official accounts, and Twitter's official blue verified badge (Yue et al., 2019). The top leaders' names, affiliations, total number of tweets, and follower information can be found in Table 1. In terms of the executives themselves, the gender disparity within this sample was 62.8% male and 37.2% female.

We designed a custom Python script that collected the 35 leaders' complete posting history through web-scraping, similar to Zhou (2021). We also collected digital public engagement data by employing the Twitter API and developing another custom Python script. The Twitter API provided information on the number of retweets and likes, while our second Python script collected the number of public comments and the comment content. Since Twitter algorithms might curate the order of a tweet's comments for logged-in users, we enabled private browsing and retrieved the top comment without user login to avoid sampling bias.

[1] To make the list, at least 10 percent of revenue must come from public support.

[2] We defined "active" twitter activity as leaders posting at least 20 tweets in the span of two years (i.e., between February 1, 2018 and February 1, 2020).

Table 1
Account information of nonprofit executives on Twitter ($N = 35$).

Executive and Twitter Name	Gender	Nonprofit Type	Company	No. of followers	No. following	Total Tweets
Sarah Hirshland (USOPC_CEO)	Female	501(c)(3)	U.S. Olympic Committee	2,698	270	127
General Brian Peddle (GeneralBPeddle)	Male	501(c)(3)	The Salvation Army	2,787	342	2,500
Alejandra Y. Castillo (A_Y_Castillo)	Female	501(c)(3)	World Young Women's Christian Association	775	196	371
Rick Evans (cxorick)	Male	501(c)(9)	New York–Presbyterian Hospital	1,926	951	2,737
Kevin Washington (KevinWash YMCA)	Male	501(c)(3)	Young Men's Christian Association	3,774	460	2,814
Aaron Kowalski (aaronjkowalski)	Male	501(c)(3)	Juvenile Diabetes Research Foundation	4,739	737	3,810
Laurie Glimcher (LGlimcherMD)	Female	501(c)(3)	Dana Farber Cancer Institute	2,393	153	435
Louis J DeGennaro (LouisJ DeGennaro)	Male	501(c)(3)	The Leukemia & Lymphoma Society	1,384	92	1,489
Sheila Musolino (RMHCCEO)	Female	501(c)(3)	Ronald McDonald House Charities	1,231	604	2,794
Michelle Nunn (MichelleNunn)	Female	501(c)(3)	CARE	11,054	533	2,335
Marta L. Tellado (MLTellado)	Female	501(c)(3)	Consumer Reports	2,266	850	424
Tamara Ruggiero (Tamara_ Ruggiero)	Female	501(c)(3)	American Kidney Fund	219	562	761
Sean Callahan (CallahanCRS)	Male	501(c)(3)	Catholic Relief Services	3,024	216	1,245
Elisa Villanueva Beard (Villanueva Beard)	Female	501(c)(3)	Teach for America	7,782	456	6,272
Jonathan Reckford (JReckford)	Male	501(c)(3)	Habitat for Humanity	2,619	79	1,984
Kaywin Feldman (Kaywin Feldman)	Female	501(c)(3)	National Gallery of Art	3,308	1,723	2,786
John Lansing (johnlansing)	Male	501(c)(3)	National Public Radio	3,255	304	596
Marc Baker (rabbimarcbaker)	Male	501(c)(3)	Combined Jewish Philanthropies	1,293	561	1,789
ohn Hewko (JohnHewko)	Male	501(c)(3)	Rotary International & The Rotary Foundation	28,811	5,663	7,386
Pam Iorio (PamIorio)	Female	501(c)(3)	Big Brothers Big Sisters of America	5,140	1,830	1,569
Cristián Samper (CristianSamper)	Male	501(c)(3)	Wildlife Conservation Society	6,904	520	7,347
Brian Gallagher (BGallagherUW)	Male	501(c)(3)	United Way Worldwide	5,597	1,159	1,411
Janti Soeripto (SaveCEO_US)	Female	501(c)(3)	Save the Children	60,490	2,096	7,380
Jon Abbott (abbottjonathanc)	Male	501(c)(3)	WGBH	591	783	1,297
Tim Shriver (TimShriver)	Male	501(c)(3)	Special Olympics	23,434	480	8,497
Allan Goodman (IIEGoodman)	Male	501(c)(3)	Institute of International Education	4,878	212	5,246
David Miliband (DMiliband)	Male	501(c)(3)	International Rescue Committee	279,071	1,372	10,548
Carter Roberts (Carter_Roberts)	Male	501(c)(3)	World Wildlife	3,470	257	841
Santiago "Jimmy" Mellado (jimmellado)	Male	501(c)(3)	Compassion	7,513	244	1,405
Adam Putnam (adamputnam)	Male	501(c)(3)	Ducks Unlimited	29,896	3,497	5,036
Avril Benoît (avrilbenoit)		501(c)(3)	Doctors Without Borders	8,554	1,687	5,297
Tim Mettey (TimMettey)	Male	501(c)(3)	Matthew 25: Ministries	509	843	5,845
Sylvia Acevedo (SylviaAcevedo)	Female	501(c)(3)	Girl Scouts	4,890	328	2,169
Joanne Miceli-Bogash (jmbogash)	Female	501(c)(3)	Cross Catholic Outreach	1,058	2,428	22,184
Paul Gionfriddo (pgionfriddo)	Male	501(c)(3)	Mental Health America	2,078	499	1,496

Note. Twitter account information retrieved in March 2020.

We examined published tweets between February 1, 2018 and February 1, 2020. In the span of two years, the 35 leaders published 24,430 tweets. To investigate original content creation, we discarded *retweets without comment* and limited our study to 14,039 tweets comprising three post categories: *original posts*, *retweets with comments (*i.e.*, quote tweets)*, and *replies*. Next, we randomly drew 20 posts from each executive, resulting in 700 qualified posts. In addition, 224 posts have received at least one comment. After deleting 29 comments that were not available [3], we analyzed the sentiment of 195 top comments.

2.6.2. Coding scheme

This study examined dialogic communication from two aspects: both functional (i.e., dialogic principles) and relational (i.e., social presence strategies) features. Dialogic principles measurement items were adopted from previous literature (i.e., Rybalko and Seltzer, 2010; Yue et al., 2019) and modified to fit the current study. In sum, dialogic principles were coded with four sub-categories: conservation of visitors, generation of return visits, dialogic loop, and information of interest to stakeholders. Executives' social presence on Twitter featured three strategies (i.e., affective, interactive, and cohesive strategies) adopted from Rourke et al. (1999) and Men et al. (2018). To evaluate executives' message tactics, this study measured message functions (i.e., information, community, and

[3] There are three scenarios where the number of comments can be correctly retrieved through web scraping but the comment text cannot be retrieved: comments deleted by users; comments originated from private accounts; comments deleted by Twitter due to content moderation.

action) from Lovejoy and Saxton (2012) with some modifications[4]. Moreover, message appeals (i.e., rational, positive, and negative appeal) were coded based on Yue et al. (2019). To operationalize public engagement, the number of likes, retweets, and comments of each post were collected, and the sentiment of comments was coded as positive, neutral, or negative (Ji et al., 2017). All items were coded as 1 being "yes" and 0 being "no." Table 2 shows the content, frequency, and intercoder reliability for each coding item.

2.6.3. Coding procedure

Two authors of the current study served as the coders, and a standardized coding book was finalized after coding training. The coding units included executives' Twitter posts and the top comment of each post. To check the intercoder reliability, 10% of randomly selected posts (n = 70) with 17 comments were coded by the two coders during the pretest. The intercoder reliability was calculated using Cohen's Kappa, ranging from 0.74 to 1.00, and was considered satisfactory (Popping, 1988) (see Table 2 for each variable's intercoder reliability scores). Inconsistencies were further discussed to ensure the coders held agreement. Each coder then coded half of the remaining Twitter posts in June 2020.

3. Results

RQ1 asks how functional (i.e., dialogic principles) and relational (i.e., social presence strategies) features of dialogic communication manifest in executives' posts. The results showed that nonprofit executives employed all four dialogic principles to some extent. Specifically, the most employed principle was "information of interest to stakeholders" (f = 587, 83.9% posts used at least one tactic), followed by "dialogic loop" (f = 547, 78.1%), "generation of return visits" (f = 486, 69.4%), and "conservation of visitors" (f = 87, 12.4%). To be more specific, executives often expressed personal opinions, thoughts, or reflections on an issue or group relevant to the nonprofit or the industry (f = 370, 52.9%). Further, a significant number of tweets featured events, activities, or services related to the nonprofit or the executive (f = 307, 43.9%) or concerned with news, reports, opinion pieces, or press releases related to the nonprofit and members (f = 294, 42.0%). Approximately one-third of these posts mentioned the nonprofit's vision, mission, and goals. In addition, executives used dialogic loop to build conversations and engagement. Tactics most often employed by executives were "tagging/@ someone/an organization to engage in a conversation" (f = 376, 53.7%) and "the use of hashtag" (f = 260, 37.2%). In comparison, less was devoted to soliciting feedback in the form of questions (f = 72, 10.3%) and surveys (f = 5, 0.7%), or inviting questions and comments (f = 32, 4.6%).

Regarding the use of social presence strategies, we found that executives build online presence by leveraging all three strategies: affective (f = 425, 60.7%), interactive (f = 502, 71.7%), and cohesive strategies (f = 515, 73.6%). Notably, they embraced emotional expressions (f = 302, 43.1%) and often disclosed their personal lives/opinions in their posts (f = 216, 30.9%). To foster mutual interactions, they made efforts to continue conversations by referring explicitly to others' messages (f = 314, 44.9%) and showed appreciation of others (f = 296, 42.3%). Furthermore, they adopted cohesive strategies by addressing publics by names (f = 376, 53.7%) and using inclusive pronouns such as *we*, *our*, and *us* (f = 218, 31.1%).

We answered RQ2 by analyzing different message functions and message appeals. Previous research shows that nonprofit organizations utilized social media to fulfill three major functions: information, community, and action (Lovejoy and Saxton, 2012; Saxton and Waters, 2014). Our results suggested that nonprofit executives were most active in disseminating information (f = 587, 83.9%) and creating an online community (f = 489, 69.9%). To a lesser extent, their messages concerned promotion, mobilization, and action (f = 312, 44.6%). Specifically, the "network-building" element of the community function outweighed the "dialogue" aspect. For instance, nonprofit executives gave recognition and thanks to stakeholders (f = 296, 42.3%) and acknowledged local events (f = 285, 40.7%) to present nonprofits as good neighbors to the community. In comparison, significantly fewer posts were "responses to reply messages" (f = 36, 5.1%) or "response solicitation" (f = 68, 9.7%). Last, for action-oriented messages, executives primarily asked followers to perform "lobbying or advocacy" (f = 272, 38.9%), followed by "promote an event" (f = 93, 13.3%). Regarding message appeals, executives predominantly applied rational (f = 533, 76.1%) and positive emotional (f = 527, 75.3%) appeals in their posts. Only a small proportion of posts (f = 55, 7.9%) contained negative emotional appeal.

RQ3 asked public engagement with executives' posts. On average, an executive's post received 25 likes (M = 24.92, SD = 80.04), seven retweets (M = 6.64, SD = 32.38), and one comment (M = 1.33, SD = 6.09). Among the 195 comments we analyzed, most comments were positive (f = 117, 60%). Neutral (f = 40, 20.5%) and negative (f = 38, 19.5%) comments each comprised approximately 20% of the total comments.

To find out how dialogic communication and message tactics affected public engagement with executives on Twitter (RQ4), we conducted a series of negative binomial regressions, controlling for executives' number of followers. We used the negative binomial regression because dependent variables (i.e., like, retweet, comment) in this study are over-dispersed count data that do not follow a normal distribution. Exponentiated coefficients [Exp(B)] represent the change in a dependent variable for every unit increase in an independent variable. For instance, an Exp(B) of 1.123 implies that for each unit change in the predictor (in our case, from *0 = no* to *1 = yes*), public engagement (i.e., like, retweet, or comment) increases by [(1.123–1) * 100%] 12.3%. A summary of regression analysis results was presented in Table 3.

The results showed that among the four dialogic principles, information of interest to stakeholders was positively correlated with all three engagement indicators: number of likes (B = 0.61, Exp(B) = 1.83, $p < .001$), retweets (B = 1.32, Exp(B) = 3.73, $p < .001$), and

[4] Information function contains a single item according to Lovejoy and Saxton's ICA framework (2012). Given its conceptual overlap with the dialogic principle of "information of interest to stakeholders," the researchers did not code information function separately.

Table 2
Descriptive information of executive posts (N = 700).

Variable	Frequency	Percentage	Intercoder reliability
Dialogic communication: dialogic principles (functional aspect)			
Conservation of visitors	87	12.4%	
Link to the nonprofit's official website	57	8.1%	1.00
Link to other social networks sites in which the executive or the nonprofit is present (e.g., LinkedIn, Facebook, YouTube, blogs, Instagram)	30	4.3%	1.00
Generation of return visits	486	69.4%	
Links to websites where additional information can be obtained	177	25.3%	0.86
Links to news/opinion pieces related to the executive or the nonprofit issued by external media/source	203	29.0%	0.89
Links to other Twitter pages: use of @	331	47.3%	0.97
Option to request information	12	1.7%	1.00
Dialogic loop	547	78.1%	
Reply by the executive to a user's post	86	12.3%	a
The provision of surveys or other channels for users to express opinions on the nonprofit	5	0.7%	1.00
The use of hashtags	260	37.1%	1.00
Request to ask a question or leave a comment (request can be done via text/video)	32	4.6%	1.00
Questions posted by the executive to solicit feedback	72	10.3%	0.87
Executive tagging /@ someone to engage in a conversation	376	53.7%	0.83
Information of interest to stakeholders	587	83.9%	
Events, activities, or services related to the nonprofit or the executive	307	43.9%	0.80
News, reports, opinion pieces, press release related to the nonprofit, nonprofit employees, or the industry the nonprofit is in	294	42.1%	0.89
Job announcements	8	1.1%	1.00
Expression of the nonprofit's vision/mission/goals	218	31.1%	0.87
Expression of the executive's personal opinions/thoughts/reflections on an issue/population relevant to the nonprofit or the industry the nonprofit is in	370	52.9%	0.80
Dialogic communication: social presence (social aspect)			
Affective strategies	425	60.7%	
Emotional expressions including descriptions, emoticons, repetitious punctuation, all-cap	302	43.1%	0.74
Humor including teasing, irony, sarcasm, joke (only if it is clearly meant to be funny)	16	2.3%	1.00
Self-disclosure, e.g., details of personal life, expressing personal opinion, including feelings of vulnerability	216	30.9%	0.90
Interactive strategies	502	71.7%	
Continue conversation; refer explicitly to others' messages	314	44.9%	0.81
Ask questions	13	1.9%	1.00
Express appreciation/compliment	296	42.3%	0.77
Express agreement	192	27.5%	0.78
Cohesive strategies	515	73.6%	
Address publics by name/@	376	53.7%	0.83
Use inclusive pronouns: address the group as *we, us, our, group, community, society, country*	218	31.1%	0.93
Phatic/salutations: communication that serves a purely social function, greetings, sharing what the executive is up to at the very moment	114	16.3%	0.84
Message function: *Community function*	489	69.9%	
Giving recognition and thanks[1]	296	42.3%	0.77
Acknowledgement of current & local events and activities	285	40.7%	0.88
Responses to reply messages (reply using the @ symbol before the name of the Twitter user the message is intended for)	36	5.1%	1.00
Response solicitation	68	9.7%	1.00
Message function: *Action function*	312	44.6%	
Promoting an event	93	13.3%	0.80
Fundraising including donation appeal & selling a product	15	2.1%	1.00
Call for volunteers	20	2.9%	1.00
Lobbying and advocacy	272	38.9%	0.77
Join another site or vote for the organization	66	9.4%	1.00
Learn how to help	25	3.6%	1.00
Message appeals			
Rational appeal	533	76.1%	0.93
Positive appeal	527	75.3%	0.78
Negative appeal	55	7.9%	0.92
Comments			
Positive comments	117	60.0%	0.96
Neutral comments	40	20.5%	0.93
Negative comments	38	19.5%	0.96

[a]Data were extracted through the Twitter Public API and a custom Python script, not manually coded.
[1]We coded the same value on "giving recognition & thanks" and "express appreciation/compliment".

comments (B = 1.08, Exp(B) = 2.95, $p < .001$). The other three principles were positively related to retweets and comments, but not likes: Dialogic loop was positively associated with the number of retweets (B = 0.43, Exp(B) = 1.53, $p < .05$) and comments (B = 0.45, Exp(B) = 1.57, $p < .01$), but not likes (B = 0.15, Exp(B) = 1.16, n.s.). Likewise, conservation of visitors was positively associated with

Table 3
Negative binomial regression results (N = 700).

Predictors	DV: Likes		DV: Retweets		DV: Comments	
	B(SE)	Exp(B)	B(SE)	Exp(B)	B(SE)	Exp(B)
Dialogic Principles						
Conservation of visitors	0.11(0.12)	1.11	0.53***(0.12)	1.70	0.46**(0.15)	1.58
Generation of return visits	0.12(0.08)	1.12	0.63***(0.09)	1.87	0.36**(0.12)	1.44
Dialogic loop	0.15(0.10)	1.16	0.43***(0.11)	1.53	0.45**(0.14)	1.57
Information of interest to stakeholders	0.61***(0.11)	1.83	1.32***(0.13)	3.73	1.08***(0.20)	2.95
Social Presence Strategies						
Affective strategy	0.58***(0.08)	1.79	0.38***(0.09)	1.47	0.93***(0.13)	2.54
Interactive strategy	-0.04(0.09)	0.96	-0.09(0.09)	0.92	0.41**(0.13)	1.50
Cohesive strategy	0.69***(0.09)	2.00	0.68***(0.10)	1.96	0.84***(0.14)	2.31
Message Functions						
Information function[1]	0.61***(0.11)	1.83	1.32***(0.13)	3.73	1.08***(0.20)	2.95
Community function	0.22**(0.09)	1.25	0.03(0.09)	1.03	0.42**(0.13)	1.53
Action function	0.06(0.08)	1.07	0.39***(0.09)	1.47	0.50***(0.11)	1.65
Message Appeals						
Rational appeal	0.49***(0.09)	1.64	0.94***(0.11)	2.55	0.97***(0.16)	2.64
Positive appeal	0.38***(0.09)	1.46	-0.02(0.10)	0.98	0.28*(0.13)	1.32
Negative appeal	-0.31*(0.15)	0.74	0.47**(0.16)	1.59	0.18(0.19)	1.20

*$p < .05$, **$p < .01$, ***$p < .001$.
Note: Each independent variable was coded as a binary variable (0 = none of the tactics were found under that principle; 1 = at least one tactic was used under the principle).
[1]*Information function* contained the same items as *information of interest to stakeholders*. We kept *information function* under *message function* and ran the regression including all three functions.

number of retweets (B = 0.53, Exp(B) = 1.70, p < .01) and comments (B = 0.46, Exp(B) = 1.58, p < .01) but not likes (B = 0.11, Exp(B) = 1.11, n.s.). Finally, generation of return visits was positively related to number of retweets (B = 0.63, Exp(B) = 1.87, p < .001) and comments (B = 0.36, Exp(B) = 1.44, p < .01) instead of likes (B = 0.12, Exp(B) = 1.12, n.s.).

In terms of the effect of social presence strategies on public engagement, we found that affective strategies positively correlated with number of likes (B = 0.58, Exp(B) = 1.79, p < .001), retweets (B = 0.38, Exp(B) = 1.47, p < .001), and comments (B = 0.93, Exp(B) = 2.54, p < .001). Cohesive strategies also showed positive correlations with the total number of likes (B = 0.69, Exp(B) = 2.00, p < .001), retweets (B = 0.68, Exp(B) = 1.96, p < .001), and comments (B = 0.84, Exp(B) = 2.31, p < .001). Interactive strategies were positively associated with number of comments (B = 0.41, Exp(B) = 1.50, p < .01) but not number of likes (B = -0.04, Exp(B) = 0.96, n.s.) or retweets (B = -0.09, Exp(B) = 0.92, n.s.).

To examine how message tactics (i.e., message functions, message appeals) affected public engagement with executives' posts, we first ran regressions with three message functions—information, community, and actions. The results showed that the information function positively affected number of likes (B = 0.61, Exp(B) = 1.83, p < .001), retweets (B = 1.32, Exp(B) = 3.73, p < .001), and comments (B = 1.08, Exp(B) = 2.95, p < .001). The community function was positively related to likes (B = 0.22, Exp(B) = 1.25, p < .01) and comments (B = 0.42, Exp(B) = 1.53, p < .01), but not retweets (B = 0.03, Exp(B) = 1.03, n.s.). Last, the action function was positively associated with retweets (B = 0.39, Exp(B) = 1.47, p < .001) and comments (B = 0.50, Exp(B) = 1.65, p < .001) but not with likes (B = 0.06, Exp(B) = 1.07, n.s.).

Regarding message appeals, rational appeal was positively related to number of likes (B = 0.49, Exp(B) = 1.64, p < .001), retweets (B = 0.94, Exp(B) = 2.55, p < .001), and comments (B = 0.97, Exp(B) = 2.64, p < .001). In comparison, positive emotional appeal was positively associated with number of likes (B = 0.38, Exp(B) = 1.46, p < .001) and comments (B = 0.28, Exp(B) = 1.32, p < .05) but not with retweets (B = -0.02, Exp(B) = 0.98, n.s.). Moreover, negative emotional appeal was positively related to retweets (B = 0.47, Exp(B) = 1.59, p < .01), negatively related to likes (B = -0.31, Exp(B) = 0.74, p < .05), and unrelated to comments (B = 0.18, Exp(B) = 1.20, n.s.).

4. Discussion and conclusions

4.1. Nonprofit executive dialogic communication on social media

This study explored dialogic communication from both a functional and relational perspective. In terms of the employment of functional dialogic communication, the overwhelming majority of tweets contained *information of interest to stakeholders,* followed by *dialogic loop* and *generation of return visits*. Although this pattern aligns with prior research, the percentage of posts featuring *information of interest to stakeholders* and *dialogic loop* were much higher (cf. Cho et al., 2014; Lovejoy and Saxton, 2012; Men et al., 2018; Saxton and Waters, 2014). When evaluating the individual items under the *information of interest to stakeholders*, it is worth noting that more than half of the examined posts are concerned with executives' personal opinions and reflections on an issue relevant to the nonprofit or its specific industry. This indicates that nonprofit leaders emphasize practicing thought leadership to reach potential stakeholders, inspire learning, and cultivate public trust in their skills and expertise (Thelen et al., 2020). Moreover, about one-third of their posts are expressions of mission, vision, and goal of their nonprofits, which implies that nonprofit executives understand the importance of using

their *personal* platform to promote the nonprofit brand and "sell" nonprofit value and purpose. Notably, this effort was less salient among prominent corporate CEOs from both established and startup companies, who, according to prior studies, did not communicate mission and vision nearly as much (Men et al., 2018; Yue et al., 2019). Interestingly, more than three-quarters of posts contained at least one dialogic loop strategy, which far exceeded the use of dialogic loop by for-profit CEOs on Facebook (Men et al., 2018). Although nonprofit executives predominantly adopted the functional, technology-afforded interactivity (i.e., more than one-third of posts used hashtag and @function) instead of openly asking questions or soliciting feedback from publics (i.e., around 10% of posts), they are, nonetheless, more proficient at generating mutual dialogues than their for-profit counterparts, who only contributed 2% of posts to questions and feedback (Men et al., 2018; Yue et al., 2019).

When it comes to the relational dimension of dialogic communication, we found that nonprofit executives capitalized on all three types of social presence strategies evenly. Furthermore, nonprofits outweighed for-profit executives in applying affective, interactive, and cohesive strategies in social media posts (e.g., Men et al., 2018; Yue et al., 2019). Almost half of the examined posts by nonprofit leaders expressed an intent to continue conversations with publics. These leaders were also more inclined to express appreciation for publics than CEOs of top Fortune and startup companies (Yue et al., 2019).

In aggregate, findings on dialogic communication seems to support a more strategic use of social media by nonprofit executives than for-profit executives. By leveraging functional and social, personal features of social media, nonprofit executives not only showed their strengths in disseminating organization-centered information and personal reflections, but also their genuineness in initiating interactions and conversations and their dedication to bringing closeness, togetherness, authenticity, and humanity to social media communication.

Results on ICA use mostly aligned with those found in previous studies; that is, nonprofit organizations prioritized providing information to stakeholders, followed by building an online community and mobilizing collective action (e.g., Huang et al., 2016; Lovejoy and Saxton, 2012; Saxton and Waters, 2014; Svensson et al., 2015; Thackeray et al., 2013). Specifically, the present study found that most tweets included at least one *informational* message, meaning top leaders were most concerned with increasing their nonprofit's visibility through one-way information-sharing. Furthermore, about two-thirds of posts served the community function, suggesting nonprofit leaders' efforts to reach out and build relationships with current or potential partners. Finally, almost half of the tweets contained at least one action-focused strategy, and among these, most tweets requested followers to perform a lobbying- or advocacy-related activity.

While action-focused messages were the least prevalent message function, the percentage reported in the current study is substantially higher than previous studies conducted at an organization-level (e.g., Lovejoy and Saxton, 2012; Guo and Saxton, 2014; Saxton and Waters, 2014; Svensson et al., 2015; Thackeray et al., 2013). This finding may indicate that leaders of nonprofits understand the role of social media as a mobilization tool and are relatively comfortable using their personal platform to solicit public support. Indeed, the ability to acquire resources and influence policy is what distinguishes effective nonprofit leaders from less effective ones (Heimovics et al., 1995).

This observation may reflect the increasing demands for organizational leaders to weigh in on social and political issues (Chatterji and Toffel, 2019) and recognize using social media for social advocacy and social movements (Xu and Xiong, 2020). Although prior literature has extensively studied the informational and dialogic features of social media communication, the *action* function deserves further attention in that it has become crucial for nonprofit organizations and leaders to spark collective actions and drive civic engagement via online mobilization (Qu, 2020).

4.2. *Effects on social media engagement*

Compared to prior studies with mixed results, our findings emphasized the overwhelmingly positive role of functional digital dialogic principles in sparking engagement. For instance, although Men et al. (2018) and Yue et al. (2020) claimed that applying *conservation of visitors* and *generation of return visits* may inversely affect engagement by driving traffic away from the social media page, the current study drew a different conclusion. Overall, all four dialogic principles engendered more retweets and comments. *Information of interest to stakeholders* is particularly effective; it not only elicited the highest retweets and comments but also generated additional benefit of "likes," which was not found with the other principles. In terms of the relational dimension of dialogic communication, this study revealed that both affective and cohesive strategies are instrumental in boosting likes, retweets, and comments. Different from prior literature, which failed to demonstrate the utility of an interactive strategy for corporate CEOs (e.g., Men et al., 2018; Yue et al., 2020), our findings showed that when a post contains an interactive component (e.g., answering questions, recognizing feedback, showing agreement), it is more likely to draw comments from online publics.

Overall, publics responded favorably to ICA messages. Notably, informational messages simultaneously elicited a greater number of likes, retweets, and comments. Although community-focused messages received more likes and comments, these messages did not increase people's sharing behavior. In comparison, action-oriented messages spawned retweets, but not likes. This finding complements a previous study that only identified publics' inclination to share informational messages (Saxton and Waters, 2014). The current study revealed that nonprofit executives can also effectively drive online sharing and word-of-mouth by centering on marketing, promotion, and calling publics to take action, whether it is signing a petition or making a donation.

In addition, we found that messages that used positive emotional appeal attracted more likes and comments, whereas messages with negative emotional appeal prompted more retweets and fewer likes. This finding reinforces a recent study that showed that posts carrying positive emotions boosted likes and comments, but not retweets (Ji et al., 2019). Complementing existing research, our findings showcased the role of positive emotional appeal from leaders in inducing interactions and relationship co-construction. As to why negative appeal increased retweeting behavior, a clearer explanation may be drawn by analyzing not only posts' emotional

valence but also types of negative emotions (e.g., anger, sadness). For instance, research shows that health-related posts carrying mostly anger were far more likely to be retweeted than those carrying sadness and disappointment (So et al., 2016). Finally, it is noteworthy that rational appeal positively affected all three engagement indicators, implying that it might be the most effective appeal for top leaders to apply.

It is also worth mentioning that among the 700 selected posts, only 224 of them had at least one comment, and on average, an executive received one comment for one post. This finding, consistent with previous research (Men et al., 2018), suggests comments as a less preferred way of public engagement with organizational leaders.

5. Implications for theory and practice

This study has made several distinct theoretical contributions at the intersection of leadership communication, nonprofit management, and social media research. First, this study advances leadership communication scholarship by illustrating the communicative and public relations function of nonprofit executives. While leadership is extensively studied as a critical determinant in the functioning and well-being of individuals and organizations, research into the communicative aspects of leadership and their public relations implications has been sparse (De Vries et al., 2010; Men et al., 2018). The role of organizational leaders is continuing to rapidly expand. This study empirically shows that nonprofit leaders are at the forefront of digital communication, trying to bridge between their organizations and stakeholders.

Relatedly, this study expands the research realm of nonprofit communication and management. Research to date has largely treated nonprofit organizations as the unit of analysis, and significantly less examined the role of nonprofit leaders. The current study presents one of the first findings showing that nonprofit executives embraced a wide array of communication objectives and strategies, such as information dissemination, community and relationship building, and advocacy and promotion, some of which greatly fostered public engagement.

Third, this study complements the growing research in social media-mediated leadership communication (e.g., Men et al., 2018; Yue et al., 2019). Political, corporate, and nonprofit leaders increasingly utilize social media to appear more direct, personal, and interactive. Social media facilitates the convergence of interpersonal and mass communication, generating exciting research opportunities surrounding mass-personal communication, featured by "publicly accessible yet highly personalized messages" (O'Sullivan and Carr, 2018, p. 1172).

In an effort to further examine this popular trend, this study integrated theoretical frameworks of different origins. For instance, dialogic communication is rooted in interpersonal communication literature with applications in organization-level communication. Following Men et al. (2018) and Yue et al. (2019), this study returned to its theoretical premise by incorporating social presence as a key relational dimension to emphasize the interpersonal nature of dialogic communication. Similarly, the ICA framework was proposed in the context of mediated organizational communication. In the current study, we extended ICA's application to explore leaders' mass-personal communication in the social media sphere. In summary, we believe that leadership communication is an important communicative situation increasingly affected by public-yet-personal mass-personal communication. We encourage future researchers to integrate mass and interpersonal communication theories to advance the theoretical development and application of leadership communication.

This study provided strategic value to executive leaders and their communication/public relations teams in the nonprofit sector. In the age of digital communication, the role of communication practitioners entails aiding top leaders in building relationships and engagement with online publics. Knowing how to select relevant topics, design effective messages, and evaluate key outtakes and outcomes is essential for creating value for organizations through communication. Furthermore, despite their relative advantages compared to for-profit CEOs, nonprofit executives should ramp up their two-way, interactive communication with the online publics. Depending on the nature of the nonprofit and the leader's characteristics, nonprofit executives can choose strategies that reflect their authenticity, openness, and expertise while satisfying publics' informational and emotional needs. In a nutshell, nonprofit executives should seize the opportunity to be the representative of their organization, engage with publics, and amplify the impact of the organization's digital presence.

6. Limitations and future research

This study presents three limitations. First, our analyses for communication strategies and public engagement outcomes were based on correlations rather than causality. Future researchers are encouraged to test the relationships in experimental settings. Second, this study used digitally embedded metrics (i.e., likes, comments, retweets) to gauge publics' interest in responding to social media posts. Future research should conduct surveys and interviews with different stakeholder groups to attain a holistic picture of publics' interpretation of leaders' social media activities. Third, we compared the results of the current study with several prior studies that adopted a similar coding scheme. These comparisons should be interpreted with caution because previous studies have looked at different social media platforms with distinct features and affordances. Additionally, scholars likely had different interpretations of the same coding items and therefore arrived at different conclusions. A more rigorous comparison of communication strategies adopted between nonprofit organizations, corporations, and government agencies can be drawn by following the same coding scheme, procedure, interpretation, and examination of the same communication platform. Last, the current study only examined top leaders of the largest nonprofits in the U.S. based on total revenue. Lessons and implications drawn from this study may not apply to those who lead small or medium nonprofits. On one hand, it is likely that smaller nonprofits would rely more on leaders' individual platforms for the purpose of engagement and relationship building than organizations' official communication platforms due to financial constraints

(Quinton and Fennemore, 2013). On the other hand, social media might level the playing field between small and large nonprofits (Elliott et al., 1998; Svensson et al., 2015). We encourage future research to further investigate and compare how leaders from small, medium, and large-sized nonprofits use social media for engagement and relationship building.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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